

The cultural environment: measuring culture with big data

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Abstract The rise of the Internet, social media, and digitized historical archives has produced a colossal amount of text-based data in recent years. While computer scientists have produced powerful new tools for automated analyses of such “big data,” they lack the theoretical direction necessary to extract meaning from them. Meanwhile, cultural sociologists have produced sophisticated theories of the social origins of meaning, but lack the methodological capacity to explore them beyond micro-levels of analysis. I propose a synthesis of these two fields that adjoins conventional qualitative methods and new techniques for automated analysis of large amounts of text in iterative fashion. First, I explain how automated text extraction methods may be used to map the contours of cultural environments. Second, I discuss the potential of automated text-classification methods to classify different types of culture such as frames, schema, or symbolic boundaries. Finally, I explain how these new tools can be combined with conventional qualitative methods to trace the evolution of such cultural elements over time. While my assessment of the integration of big data and cultural sociology is optimistic, my conclusion highlights several challenges in implementing this agenda. These include a lack of information about the social context in which texts are produced, the construction of reliable coding schemes that can be automated algorithmically, and the relatively high entry costs for cultural sociologists who wish to develop the technical expertise currently necessary to work with big data.

Keywords Culture · Content analysis · Mixed-methods · Evolutionary theory

More data were accumulated in 2002 than all previous years of human history combined.¹ By 2011, the amount of data collected prior to 2002 was being collected every 2 days.² This dramatic growth in data spans nearly every part of our lives—from

¹International Data Corporation, “The 2011 Digital Universe Study: Extracting Value from Chaos,” June, 2011. See also Christopher R. Johnson, “How Big is Big Data?” Lecture at the University of Michigan’s Cyber-Infrastructure Conference, November 7th, 2012.

²Ibid.

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gene sequencing to consumer behavior.³ While most of these data are binary or quantitative, text-based data are also being accumulated on an unprecedented scale. In an era of social science research plagued by declining survey response rates and concerns about the generalizability of qualitative research, these data hold considerable potential (Golder and Macy 2011; King 2011; Lazer et al. 2009). Yet social scientists—and cultural sociologists in particular—have largely ignored the promise of so-called “big data.” Instead, cultural sociologists have left this wellspring of information about the arguments, worldviews, or values of hundreds of millions of people from Internet sites and other digitized texts to computer scientists who possess the technological expertise to extract and manage such data but lack the theoretical direction to interpret their meaning.

The most obvious explosion in text-based data coincided with the rise of the Internet. Between 1995 and 2008 the number of websites expanded by a factor of more than 66 million, recently surpassing 1 trillion.⁴ Although sociologists were understandably concerned about digital divides in years past, these inequalities appear to be steadily decreasing (DiMaggio and Bonikowski 2008; Dimaggio et al. 2001). According to a 2012 survey, roughly half of all Americans visit a social media site such as Facebook or Twitter each day, producing billions of lines of text in so doing.⁵ These trends are markedly higher among younger people, suggesting these trends may only continue to grow over time.⁶ Most of the text from social media sites is readily accessible via simple computer programs.⁷ Yet the outgrowth in text-based data on the Internet is not limited to social media sites. Screen-scraping technologies can be used to extract information from any number of Internet sites within time frames that are only limited by digital storage capacity.⁸ And the potential to collect such data is not only tied to the future, but also the past. Since 1996, a non-profit organization known as the Internet Archive has been storing all text from nearly every website on the Internet.

The outgrowth of text-based data is also not confined to the Internet. Thanks to new digital technologies from fields as diverse as library science and communications, an unprecedented amount of qualitative data is being archived. Google alone has already created digital copies of nearly every single book ever written in collaboration with more than 19 million libraries worldwide.⁹ Academic data warehouses such as LEXIS-NEXIS or ProQuest now contain digital copies of most of the world’s journals, newspapers, and magazines. The Vanderbilt Television News Archive contains copies of most major

³ The US National Science Foundation invested more than \$15 million in Big Data projects in 2012, and will easily surpass this amount in upcoming years due to the development of new infrastructure for funding big data projects in collaboration with Britain’s Economic & Social Research Council, the Netherlands Organization for Scientific Research, and the Canada Foundation for Innovation, among many others.

⁴ Jesse Alpert and Nissan Hajaj, “We knew the web was big...” Official Google Blog, July 25th, 2008 (<http://googleblog.blogspot.com/2008/07/we-knew-web-was-big.html> accessed January 2012).

⁵ Pew Internet & American Life Project, February 1st, 2012.

⁶ “Social Networking Popular Across Globe,” Pew Research Global Attitudes Project, December 12, 2012.

⁷ Moreover, the US Library of Congress recently announced plans to release a database of every single Twitter message ever made. Current estimates place the total number of tweets that might be archived at more than 170 billion.

⁸ Web-scraping technologies have facilitated the collection of remarkably large datasets. Golder and Macy (2011), for example, recently conducted a study of more than 500 million Twitter messages produced in more than 84 countries over a 2 year period.

⁹ Though access to the entire Google book archive is limited by pay walls designed to protect copyright privileges, Google has released the entire dataset in “ngram” format, which allows scholars to analyze them via the automated text analysis tools discussed in further detail below.

newscasts produced since 1998. An unprecedented amount of text-based data that describe legislative debates, government reports, and other state discourse is also now available on websites such as the National Archives of the United States and Great Britain. Qualitative academic research is also being compiled within “meta-data” archives on an unprecedented scale—from in-depth interview data to field notes.¹⁰ Continuing improvement in digital speech recognition technologies has also facilitated even more text-based data from historical audio sources to local town hall meetings that are recorded and uploaded to websites for posterity. Indeed, the remarkable growth in text-based data warrants a brief thought experiment: what types of text or speech-based data are *not* currently being archived?

If the answer is little—or very little—text is not being archived, then cultural sociology must have a reckoning with big data alongside those in other fields.¹¹ Political scientists are currently exploring the potential of social media to explain political mobilization (Hopkins and King 2010; Livne et al. 2011). Public health scholars use Twitter to identify trends in disease (Paul and Dredze 2011), and communications scholars claim it can be used to predict shifts within the stock market (Bollen et al. 2011). Even humanities scholars have invented the vibrant new field of digital humanities (e.g. Gold 2012; Moretti 2013; Tangherlini and Leonard 2013). By comparison, cultural sociologists have made very few ventures into the universe of big data even though texts are a central object of study in the field—in the form of primary documents, interview transcriptions, or field notes.¹² In this article, I argue inattention to big data among cultural sociologists is doubly surprising since it is naturally occurring—unlike survey research or cross-sectional qualitative interviews—and therefore critical to understanding the evolution of meaning structures *in situ*. That is, many archived texts are the product of conversations between individuals, groups, or organizations instead of responses to questions created by researchers who usually have only post-hoc intuition about the relevant factors in meaning-making—much less how cultural evolves in “real time.”¹³

For all the promise of big data for cultural sociology, formidable obstacles remain. First of all, the sheer volume of data can be overwhelming. Large corpora cannot be coded by hand, and automated data mining techniques are of little utility if they are not guided by theory. Second, big data is untidy. Although computer-assisted data classification and data reduction techniques have improved in the past decade, much big data analysis remains computationally intensive and therefore out of reach for many cultural sociologists—particularly those without any background in statistics or computer programming. Third—and perhaps most importantly—there is much that is of interest to cultural sociologists that is not easily reducible to text. The greatest challenge for cultural sociologists interested in big data is to develop new techniques to measure the unspoken or implicit meanings that occur in-between words. The preconscious cultural scripts or frames that shape how people understand the world (e.g., DiMaggio 1997),

¹⁰ See, for example, the Dataverse Network, the Interdisciplinary Consortium for Political and Social Research, and the United Kingdom’s Qualidata archive.

¹¹ The neologism “big data” has come to refer to many different types of data. Here, I use the term to refer to the increasingly large volume of text-based data that is often-though not always- produced through digital sources. As the remainder of this manuscript describes, these data are also unique because they are “naturally occurring,” unlike survey data which result from the intrusion of researchers into everyday life.

¹² Exceptions described in additional detail below include Franzosi (2004), Lewis et al. (2008), Bail (2012), Bail (forthcoming) and several other works in progress.

¹³ “Real time” refers to the collection, presentation, or analysis of data at or very near the time it is being produced by social actors.

for example, are not always manifest in speech or text. Similarly, most big data eschews the production of meaning through bodily interaction (e.g., Eliasoph and Lichterman 2003)—though the future of big data may include new techniques to analyze the ever-increasing volume of video on the Internet (Collins 2013; Lan and Raptis 2013).

This article does not offer solutions to each of these limitations of big data. Instead, it provides a critical survey of recent developments within the big data movement and links them to outstanding theoretical debates and measurement challenges within cultural sociology. These include the measurement of cultural environments or meaning systems such as discursive fields; the classification of cultural elements such as frames or schema within such systems; and tracing cultural processes over long segments of time. In describing the promise of big data for cultural sociology, I also detail how the latter field may address some of the most vexing challenges of the former given its foundational interest in the systematic study of meaning. I provide only limited discussion of the technical and logistical issues that arise in working with big data because these issues are currently being addressed within separate literatures referenced below.¹⁴ I also do not review the promising field of quantitative narrative analysis because it has been addressed elsewhere.¹⁵ This article is thus an invitation to cultural sociologists curious about the potential of big data and a call to shatter the disciplinary silos that inhibit collaboration between this field and those who lead the big data movement.

Mapping cultural environments

By and large, the central objects of study in cultural sociology have been confined to micro-levels of analysis. For example, cultural elements such as symbolic boundaries (e.g., Lamont 1992), cultural toolkits (e.g., Swidler 1986), cognitive schemas (e.g., DiMaggio 1997), and cultural frames (e.g., Benford and Snow 2003) have been defined as judgments, classifications, or pre-conscious decisions that can only be measured through close readings of texts such as interview transcripts, content analysis of key texts, or ethnographic field notes. Yet as Swidler (1995) argues, “the greatest unanswered question in the sociology of culture is whether and how some cultural elements control, anchor, or organize others.”¹⁶ For example, how are cultural frames ordered within vast discursive fields? Is there a space between such fields? How do cultural frames shape the evolution of fields more broadly? Addressing such questions requires meso- and macro-level analysis of the relationship between multiple cultural elements or systems of meaning. One of the most promising dimensions of the big data movement for cultural sociology is to enable new analyses at these larger levels of analysis. As I describe below, one can now obtain every website, blog, social media message, newspaper article, or television transcript on a given topic fairly easily.

The capacity to capture all—or nearly all—relevant text on a given topic opens exciting new lines of meso- and macro-level inquiry into what I call cultural

¹⁴ For a technical overview of techniques designed for analysis of Big Data, see Manning and Schuetze (1999).

¹⁵ For an overview, see Franzosi (2009).

¹⁶ See also Ghaziani and Baldassarri (2011).

environments (Bail [forthcoming](#)). Ecological or functionalist interpretations of culture have been unpopular with cultural sociologists for some time—most likely because the subfield defined itself as an alternative to the general theory proposed by Talcott Parsons (Alexander [2006](#)). Yet many cultural sociologists also draw inspiration from Mary Douglas (e.g., Alexander [2006](#); Lamont [1992](#); Zelizer [1985](#)), who—like Swidler—insists upon the need for our subfield to engage broader levels of analysis. “For sociology to accept that no functionalist arguments work,” writes Douglas ([1986](#), p. 43), “is like cutting off one’s nose to spite one’s face.” To be fair, cultural sociologists have recently made several programmatic statements about the need to engage functional or ecological theories of culture. Abbott ([1995](#)), for example, explains the formation of boundaries between professional fields as the result of an evolutionary process. Similarly, Lieberman ([2000](#)), presents an ecological model of fashion trends in child-naming practices. In a review essay, Kaufman ([2004](#)) describes such ecological approaches to cultural sociology as one of the three most promising directions for the future of the subfield.¹⁷

The concept of discursive fields is perhaps the most promising theoretical construct to advance an ecological approach to cultural sociology (Bourdieu [1975](#); Foucault [1970](#); Martin [2003](#); Wuthnow [1993](#)). Yet field theory is often castigated for being tautological, or assuming the existence of invisible or intangible social forces that reproduce structures of inequality or patterns of cultural differentiation without ever directly observing them. The boundaries of fields are usually unobserved in empirical studies because of the considerable methodological obstacles involved. Apart from Eyal ([2009](#)), cultural sociologists have scarcely theorized the outer limits of cultural fields, the spaces between them, or the relationships among multiple fields.¹⁸ This is a significant limitation since most field theory makes several assumptions that are inherently ecological. For example, many studies assume that relationships between actors or groups of actors within a field produce a polarity that sustains or reproduces uneven power relationships or access to institutions (Bourdieu [1985](#); Fligstein and McAdam [2011](#); Wuthnow [1993](#)). Others borrow more directly from ecological or evolutionary theory to explain the competition for attention or resources within fields (Abbott [2001](#); Kaufman [2004](#); Lieberman [2000](#)), or the ability of cultural entrepreneurs to exploit niches within such environments (e.g., Mische [2008](#)).

Despite the implicit ecological reasoning of field theory, most applications of this framework rely upon micro- or meso-level measurement strategies. For example, many studies identify key actors or institutions within fields and trace their influence over other parts of the fields. Other studies focus upon conflict or “classification struggles” within fields in order to identify such influential actors (Bourdieu [1990](#)). As a result, these types of studies only observe the consequences of field-level processes rather than meso- or macro-level relationships between social actors and cultural elements that most scholars believe create such social spaces.¹⁹ These micro-level measurement strategies are typically necessary because most discursive fields are so broad that an entire team of researchers working for several years could only map a fraction of all the texts, transcripts, or archives that define them. The size of most cultural fields has become even more daunting with the rise of the Internet. Indeed, a researcher could

¹⁷ See also Mark ([2003](#)).

¹⁸ One exception is Evans and Kay’s ([2008](#)) study of field overlap.

¹⁹ Exceptions include Mohr and Guerra-Pearson ([2010](#)) and Bail ([2012](#)).

easily follow links between websites for hours only to forget where, when, or why they shifted focus from one site to another.

The big data movement has made extracting all text from a discursive field easier than ever before.²⁰ Massive databases already exist that classify texts into meaningful social categories. For example, services such as LEXIS-NEXIS and Pro-Quest have sophisticated searchable indexes that cover industries, geographical location, time, or different types of text (e.g., newspapers, newswires, or television transcripts). Simple Boolean operates such as “AND” and “OR” can be used to further specify meaningful cultural environments within each of these sub-samples.²¹ Yet perhaps the most powerful innovation of the big data movement for the mapping of cultural environments has been screen scraping, or automated extraction of text from websites. Screen scraping is typically used to mine text or other data from web pages, though it can also be used to extract text from scanned images using Optical Character Recognition (OCR) technologies. A variety of data archives have developed searchable indexes based on such screen-scraping technologies. Google, for example, allows Boolean searches of its archives of books, blogs, government documents, and major US newspapers and magazines.

But new technologies produced by the big data movement have also advanced automated extraction of text far beyond simple indexes, Boolean searches, and screen scraping. In particular, new techniques have been developed to exploit the relational nature of many sources of big data—particularly those from the Internet. For example, Gong (2011) recently introduced new software that fuses snowball sampling methods with screen-scraping technologies. The user simply inputs a starting website and a classifying rule such as a Boolean search term or one of the other classification algorithms described in further detail below. The software then visits each site that is linked to the starting website and uses the classifying rule to decide whether it should be included in the sample. If so, the program extracts all text from the site and repeats the process of “spidering” links across multiple waves that are only constrained by computer memory processing power. Given a number of different starting sites and a sufficient number of waves, the Snowcrawl software produces a *total sample* of all websites pertaining to a given topic. Although this tool is currently limited to the Internet, a number of other qualitative data archives store relational data that could potentially be analyzed using similar automated snowball methods. What is more, the majority of newspapers, television stations, journals, or other texts of interest to cultural sociologists are now available on the web.

A second promising tool for extracting large amounts of data from the web or qualitative data archives are Application Programming Interfaces (APIs). These web-based tools provide an interactive interface with large data archives that are designed to enable targeted data extraction. They were developed primarily for consumer purposes—such as the creation of third party applications for social media sites such as

²⁰ While automated data extraction methods are particularly useful for mapping the contours of discursive fields, it is important to note that such techniques do not capture the deeper preconscious cultural elements that undergird social fields as Bourdieu and others have theorized them (e.g., Bourdieu 1990; Fligstein and McAdam 2011; Martin 2003). I return to the question of whether big data techniques can be leveraged to classify such cultural elements in the following section as well as my discussion and conclusion.

²¹ For example, one might define a discursive field by identifying all texts with a certain set of keywords or within a certain search index offered by text archives.

Facebook, Twitter, or Google—but a number of academics have begun to use them as data collection tools as well (Bail 2013a; Gaby and Caren 2012; Livne et al. 2011). Even conventional media outlets such as the *New York Times* now offer APIs that enable users to search and download articles or user comments from their website. APIs are superior to other forms of data extraction not only because they enable more sophisticated targeting of different types of text—such as Twitter messages about the Arab Spring—but also because such sites typically record a vast array of information about the users of their sites as well as their behavior online. For example, Twitter’s API enables rapid extraction of information about the online social networks of individual users. Facebook and Google’s API enable direct interface with its massive archives of web content as well, but also includes information about the size, geographic location, and demographic characteristics of the audiences of each site as well.²²

Classifying culture

Obtaining total or near total samples of text on a given topic is a remarkable feat given that it was nearly unthinkable only a decade ago. Yet such giant samples are of little utility if they cannot be classified in a meaningful manner. Cultural sociology has been fascinated with classification since its inception because it was largely inspired by the Durkheimian idea of classification struggles (e.g., Barth 1969; Bourdieu 1975; Douglas 1966; Latour 1988). For example, Gieryn (1999) highlights the critical role of social classification in the evolution of scientific fields. Lamont (1992, 2000) explains how class and racial boundaries shape the process of group formation. Finally, Espeland and Stevens (1998) make a broader argument about the key role of commensuration in producing social power.²³ Yet for all the theoretical interest in the process of classification, cultural sociologists seldom discuss the appropriate way to measure social categories (Lamont and White 2009). Most studies either rely upon in-depth interviews or case studies that highlight the social construction of ranking within institutions. The lack of consensus about how to classify data has even prompted some critics to accuse cultural sociologists of the reification of social classifications according to their theoretical persuasion (e.g., Biernacki 2012).

To date, cultural sociologists have scarcely explored the promise of automated text analysis to classify texts.²⁴ Where these techniques have been used they have been relatively primitive approaches to automation that simply identify keywords or phrases. This approach is severely limited because it requires the researcher to have an a priori sense of which terms are well suited to address the theoretical question of interest. Moreover, it eschews the broader context of words within sentences. One solution to this problem is to evaluate the co-prevalence of words within sentences using Global Regular Expression Print (GREP) commands available in qualitative software analysis programs such as Atlas.TI or

²² Facebook’s API requires user-authentication to access these data. Therefore, one must either access only publicly available data or obtain an authentication token from a Facebook page’s owner. Elsewhere, I argue that app-based technologies are the most promising data collection tools to overcome such challenges. See Bail (2013b).

²³ For a recent review of this literature, see Lamont (2012).

²⁴ Notable exceptions discussed in further detail below include Mohr (1998), Franzosi (2004), Bearman et al. (1999), Bearman and Stovel (2000), Smith (2007), and Bail (2012).

WordStat. Yet these approaches nevertheless fail to recognize important nuances in the use of language. For example, a GREP search for sentences with the terms “President” and “hate” would reveal both “I hate the President,” and “I’d hate to be President.”

Recent technological advances within the fields of computer science, pattern identification, and linguistics have produced a variety of superior alternatives. I begin by reviewing “unsupervised” text classification techniques that rely exclusively on computer algorithms to create meaningful groupings of texts. For example, recent studies have invoked a number of different forms of multi-dimensional scaling or cluster analysis to classify texts (e.g., Grimmer and King 2011; Livne et al. 2011).²⁵ These techniques replace each unique word in a document with a number and then use various metrics to calculate dissimilarities among all texts in the sample. These measures may be plotted within multidimensional space in order to identify meaningful groupings of documents. A substantial problem with cluster analysis is that the results are highly sensitive to the researcher’s assumptions about the number of possible clusters (k), as well as the mathematical distances employed within each algorithm. These idiosyncrasies can be controlled, however, if multiple forms of cluster analysis are used in tandem. Grimmer and King (2011), for example, have developed software that applies all existing variants of cluster analysis to large text corpora. They apply this powerful tool to thousands of political texts by or about US presidents in order to classify their ideological position on a range of substantive issues.

Another promising development within the big data movement for cultural sociologists is the burgeoning field of “machine learning” and specifically the field of topic modeling. This new field resulted from collaboration between linguists and computer scientists designed to identify hidden or latent themes within large corpora.²⁶ Topic Models identify such themes using probabilistic models that evaluate the co-occurrence of words. The most popular form of topic modeling is Latent Dirichlet Allocation (LDA), which assumes a random allocation of words across a latent theme or topic and then uses a generative process of classification to analyze the probability of a document containing information about a topic given the distribution of words therein.²⁷ Dozens of studies have used LDA or related Bayesian approaches to infer latent topics in scientific journals, news articles, or blog posts (e.g., Blei and Lafferty 2007; Hopkins and King 2010; Quinn et al. 2010). Despite these advances, topic models have several considerable limitations. For example, the method assumes that the order of words in a document does not matter, as well as the order of documents within the broader sample. Most topic models also required that each document be assigned to mutually exclusive categories, and do not recognize relationships between topics themselves. Basic topic models also do not recognize that topics may shift or combine over time. Finally, topic models—not unlike cluster analysis—must be validated in order to verify the appropriate number of topics within a corpus.²⁸ This is particularly difficult given that many cultural sociologists are interested in analyzing broad, unstructured samples of text such as those described in the previous section of this article.

²⁵ Mohr (1998) made early calls for cultural sociologists to adopt these methods to classify meaning structures, yet they were mostly ignored even as they become widely used by cognitive anthropologists (e.g., D’Andrade 1995).

²⁶ For an overview of this field, see Blei (2012).

²⁷ For a technical overview of LDA, see Blei et al. (2003).

²⁸ A number of scholars have proposed validity measures for LDA, most recently Blei (2012). Most of these emphasize comparisons of topic models via log-likelihoods or harmonic means, yet most proponents of topic modeling agree that they must also be validated via qualitative inspection of individual topics within subsets of large samples.

Proponents of topic modeling have already begun to develop a number of solutions to these limitations of this method, though they are too technical to discuss here.²⁹ Among the more promising recent developments in the field is the advent of “supervised” topic modeling (Blei and McAuliffe 2010). In this technique, a human coder identifies topics within a subset of documents, and topic models use these assignments to assess probability instead of assuming that the distribution of topics across documents is random. Supervised text classification was first introduced within social science by Hopkins and King (2010), who used this approach to assess public opinion of presidential candidates expressed upon thousands of political blogs during the 2008 election.³⁰ Given a sufficient number of “training documents” produced through in-depth coding, these authors argue that their technique classifies sentiment about presidential candidates more reliably than human coders themselves.³¹ While such claims have not yet been widely validated, supervised learning techniques hold considerable promise for the purpose of identifying cultural elements within texts and further improving the snowball sampling methods described above.³²

Perhaps the most important question for cultural sociologists interested in employing topic models is whether they can be used to classify cultural elements such as frames, symbolic boundaries, or cultural toolkits. A number of current studies suggest topic models may be used to capture such nuanced cultural elements. For example, Dimaggio et al. (forthcoming) argue topic models can be used to identify frames about arts funding. Polletta is currently using topic modeling to identify hidden frames in Internet discussions about cap-and-trade.³³ Hopkins (2013) employs topic models to measure frames about the Affordable Care Act. Yet a key issue remains whether cultural frames—as Goffman (1974) first defined them—can be represented by groups of words. While the face-work that Goffman emphasized is clearly not measurable through text, Goffman himself used texts extensively throughout his work, including biographies, newspaper clippings, and transcripts of interactions.³⁴ Although Goffman emphasized the absence of certain words as much as the presence of others—these omissions could be modeled effectively because they would shape the probability distributions around groups of words that LDA analyzes to create classifications of texts. Nevertheless, the quality of supervised topic modeling is only as good as the codes developed by human coders themselves. Therefore, cultural elements that are highly nuanced or situation-based are not easily captured via this technique because of low inter-coder reliability.³⁵

²⁹ For example, see Blei and Lafferty (2006), Wallach (2006), Chang et al. (2009), and Hopkins and King (2010).

³⁰ See also Grimmer (2010) and Quinn et al. (2010).

³¹ In particular, Hopkins and King (2010) argue that coding more than 500 documents produces diminishing returns in the reliability of automated text analysis.

³² For example, a supervised topic model can be used to determine whether websites should be included in a directed web-crawl such as SnowCrawl to capture sites that discuss a theme or topic without using a single key-word.

³³ See Baumer et al. (2013).

³⁴ Consider, for example, the diaries analyzed in Goffman (1963) or the newspaper clippings in Goffman (1974). Also, textual descriptions of face-work or other unspoken forms of bodily interaction in the form of field notes could potentially be analyzed using topic models.

³⁵ For a discussion of the challenges of achieving high levels of inter-coder reliability in cultural analysis, see Krippendorff (2003).

On the other hand, the meticulous coding definitions required by topic models may also provide an opportunity for cultural sociologists to contribute new methodologies to the big data movement. Indeed, the use of generative and multi-stage coding schemes has been a key concern of cultural sociology in the form of “thick description” (e.g., Geertz 1973), “middle-range theory” (Merton 1949), “structural hermeneutics” (Alexander and Smith 2001), and “paradigmatic clusters” (Weber 2005). Each of these approaches emphasizes that researches should move back and forth between different levels of analysis to tune their coding schemes and to assess the scope conditions of a particular finding. To this end, the expertise of cultural sociologists may be applied to repeated stages of supervised topic models—elaborating classification systems as if they were Russian Dolls, to borrow Bourdieu’s metaphor. Mohr et al. (2014), for example, have advanced this technique in their study of US National Security Statements over a 22 year period. By developing increasingly precise codes from iterative qualitative analysis of small sub-sets of this large corpus of text, these scholars have developed increasingly promising topic models that can later be applied to the entire sample. We need further empirical validation of such techniques. At the very least, however, such methods provide a systematic method of focusing qualitative microscopes within the increasingly overwhelming world of big data.

Tracing the evolution of cultural environments

One of the most promising elements of the big data movement is that so much of the qualitative data that has been collected is longitudinal. For example, the Library of Congress’s archive of all Twitter messages will enable unprecedented analysis of how different issues rise and fall over time. The Internet Archive and screen-scraping technologies could be used to map shifts in the discourses of different types of websites over time. Likewise, the massive newspaper and television transcript archives now available could be used to analyze similar issues over the past century. These longitudinal data are particularly promising because so many of the most pressing questions in cultural sociology concern change over time. While Swidler’s (1986) toolkit analogy has received extensive attention in recent decades, for example, her call for future studies to examine the transition from unsettled to settled historical periods has been mostly ignored.³⁶ While Sewell’s (1996) theory of events has inspired considerable interest, few studies place such events in broader historical context.³⁷ Finally, Lamont’s (1992) work reveals considerable cross-national differences in the salience of symbolic boundaries. Yet we urgently need broad historical analyses to identify how such divergent meaning systems evolved over time. Each of these outstanding questions requires methods capable of capturing broad-scale cultural change.

In addition to identifying cultural elements such as frames or symbolic boundaries, automated text analysis can be used to differentiate social actors or key events within

³⁶ But see Cerulo (1998), Wagner-Pacifici (2010), and Bail (2012).

³⁷ Still, historical analyses with big data are limited by the availability of texts produced during this period that were amenable to digitization. This presents a number of important limitations, including pervasive illiteracy during early historical periods as well as the tendency for only elite accounts of historical events to survive the passage of time. Still, comparative-historical sociologists face these problems regardless of whether they are working with big data. Furthermore, primary documents obtained through archival analysis can be easily digitized through photographs, scanning, and text-recognition technologies.

large qualitative datasets.³⁸ Cultural sociologists can make huge strides towards advancing theories of social change simply by mapping the relations among cultural elements, actors, and events over time. The literature on quantitative narrative analysis has already established how analysis of relationships between actors and events can be used to map broad historical sequences (e.g., Bearman et al. 1999; Bearman and Stovel 2000; Franzosi 2004; Smith 2007). Incorporating cultural elements identified via topic modeling into such methods would open exciting new lines of inquiry about the interpenetration of culture and structure. If topic modeling can be used to identify actors and organizations as well as the cultural elements they produce, for example, social network relationships might be mapped onto cultural patterns—or vice versa. At a minimum, mapping the relationships among cultural elements, actors, and events would help focus in-depth qualitative analysis of key historical shifts or “turning points” (Abbott 1997) where meaning structures change.³⁹

One problem, of course, is that cultural elements themselves often change throughout such broad-scale historical transformations. Sewell, for example, argues the very concept of revolution was developing at the same time that murderous mobs stormed the Bastille—setting off the French revolution, before they knew precisely what they were doing. Topic models are ill equipped to capture such nuances unless human coders calibrate them repeatedly across multiple time periods. Even then, slight shifts in cultural elements may be difficult to code automatically because human coders may struggle to achieve high inter-coder reliability. Here again, new tools for automated text may prove useful. For example, several new methods have been developed to identify dissimilarities between pairs of documents. Primitive forms of these techniques simply count the number of words shared between the two documents. Yet recent advances in plagiarism detection software employ “word-maps” that utilize data from thesaurus in order to identify “near” matches between two documents as well (e.g., Bail 2012). Once again, these document comparison tools will not identify cultural elements by themselves. Yet they may be particularly powerful when combined with topic models and micro-level qualitative analysis of key texts or transitional moments within history.

Another major advantage of big data is that much of it includes detailed information about relationships between social actors. This is particularly true of social media sites such as Twitter or Facebook, but advances in library science are also creating hyperlinks between texts within archival collections as well. Using Twitter’s Application Programming Interface, one can easily extract not only all the messages produced by a single actor, but also the precise location of this actor within a broader social network—including measures of both “in” and “out” degree. Livne et al. (2011) for example, extracted 460,000 tweets from all candidates for US House, Senate, and Gubernatorial elections between 2006 and 2010. Their data not only reveals the partisan networks of such social actors, but also patterns in the similarity of the language they post on Twitter via cluster analysis. Through this analysis, Livne et al. document the meteoric rise of the

³⁸ If key actors or events are already known, simple key word searches or Global Regular Expression Print (GREP) commands may also be used to identify them. If actors or events are not known, they can be identified through keyword counts that remove common words such as “the” or “and.” Once actors or events are defined, topic models may be used to identify them as well. A number of computer scripts have also been recently developed to identify names within big data without such intermediary steps such as the Natural Language Toolkit and the Stanford Parser.

³⁹ See also Sewell (1996) and Wagner-Pacifici (2010).

Tea Party in recent elections, and the realignment of mainstream conservative networks that ensued. These and other datasets could be used to address a number of key questions at the intersection of cultural sociology and network theory. For instance, Pachucki and Breiger's (2010) argument about "cultural holes" within networks, or Vaisey and Lizardo's (2010) theory that cultural worldviews influence network composition.⁴⁰

The potential to assemble large datasets that describe cultural elements, actors, events, and social networks over time may also encourage critical advances in field theory. Most of the most pressing questions in this literature are about the evolution of fields over time (Fligstein and McAdam 2011; Padgett and Powell 2012). For instance, a number of recent studies have begun to analyze the emergence of fields (e.g., Armstrong 2002; Bartley 2007). By and large, these case studies are unable to investigate a variety of broad cultural processes that may occur between discursive fields. For example, do most fields emerge out of the dissolution of others? Or, do fields develop when the space between any two pre-existing fields is sufficiently broad (Eyal 2009; Medvetz 2012)? Big data may also enable analysis of a number of intriguing questions within individual fields as well. For example, do discursive fields have carrying capacities for new forms of culture? Do certain actors gain power within discursive fields by exploiting niches between rival factions? Or, what is the relationship between the core and periphery of discursive fields (e.g., Bail 2012)?

Another exciting feature of big data is that it often includes geo-coded data. For example, Twitter and Facebook record the geographic location of their users. This information is also often recorded on the comments sections of websites. Finally, analytics or "insights" data often include the latitude and longitude of visitors to different websites via Internet Protocol (IP) addresses or other geographic identifiers such as city names. Political scientists have even mined visual data on ethnic conflict from Google Earth (Agnew et al. 2008). The potential to look at the relationship between Cartesian coordinates and cultural elements could create a new subfield within cultural sociology that analyzes the geography of meaning. Such a field might examine questions such as: 1) Do cultural frames or symbolic boundaries cluster at the national level or supranational levels? 2) Does physical proximity breed more convergence of worldviews than online interaction? Finally, does the answer to these two questions change over time as the forces of globalization push people ever closer together?

Conclusion

Cultural sociology has long suffered from an imbalance of theory and data (Ghaziani 2009). Yet the big data movement may radically alter this equilibrium. The big data movement began with the Internet and social media, but the future of the field will also entail increasingly ambitious forays into the past. As digitized historical archives continue to expand and social scientists coordinate new ways of organizing qualitative meta data with rich detail about the evolution of meaning, cultural sociologists can no longer afford to ignore the big data movement. Above, I argued that integration of in-depth qualitative coding techniques pioneered by cultural sociologists and anthropologists can be leveraged to improve already powerful automated text analysis techniques

⁴⁰ On the concept of cultural holes, see also Lizardo (in this issue).

produced by computer scientists, linguists, and political scientists. This synthesis will enable cultural sociologists to achieve theoretical progress on questions that were once thought un-measurable. Proponents of big data may also gain key insight from cultural sociologists about how to further hone their tools to map the contours of cultural fields, classify cultural elements, and trace the evolution of culture over time.

Yet for all of my optimism about the marriage of cultural sociology and big data, formidable obstacles remain. Perhaps the most vexing problem is that big data often does not include information about the social context in which texts are produced (Griswold and Wright 2004). Although we are able to collect millions of blog posts about virtually any issue, these data typically include little or no information about the authors of such posts—or those who comment upon them. Twitter data are publicly available, but provide very little information about the social context in which Tweets are produced. Other sites such as Facebook collect massive amounts of data about social context but are often unable to share them with researchers because of concerns about user privacy. Sources of big data outside social media also often lack important information about the social context in which texts are produced. Collecting every newspaper article on a political topic is of marginal utility absent in-depth analysis of the political and institutional processes that lead media to gravitate towards one issue over another.⁴¹ Yet these obstacles are not without solutions that might build upon the progress of cultural sociologists in developing mixed-method research designs. For example, qualitative or quantitative surveys of Twitter users could be conducted to place their online behavior within broader context. Or, large-scale analyses of media data or historical surveys might be used to identify compelling puzzles for comparative historical analysis. In theory, big data could also be used to guide ethnographic interventions as well—or at least help place the findings of ethnography within broader cultural fields. *In brief, big data methods should be viewed as a complement—not a replacement—for the tried and tested techniques of cultural sociology.*

A second major challenge is that computer-assisted coding can never be more reliable than the codes themselves. Cultural sociologists seldom discuss coding criteria or inter-coder reliability, in part because the definition of many of our core concepts is highly contested (Biernacki 2012). One need only read the literature on framing, for example, to witness significant disagreement about whether and how they should be measured or operationalized.⁴² While these debates will not be easily resolved, the integration of big data and cultural sociology will depend critically upon our capacity to converge upon several broadly accepted definitions of these core concepts. Yet big data may actually facilitate such conversations—since conceptual vagueness among cultural sociologists results in part from our paucity of shared datasets. Cultural sociologists are also looking across disciplinary lines for guidance in making core concepts more concrete. For example, Mohr et al. (2013) have fused the literatures on narrative from

⁴¹ It is also worth noting that texts that cannot be collected because they are not in the public domain may ultimately have less impact upon the evolution of broader cultural domains precisely because they are hidden from public view. This underlies a broader pragmatist argument about the need to focus attention upon consequences of social action (e.g., Johnson-Hanks et al. 2011; Tavory and Timmermans 2013). An interesting analogue is the debate about the social construction of ethnicity via the enumeration of different groups by the US Census (cf. Loveman and Muniz 2007). I thank Andy Perrin for bringing this issue to my attention.

⁴² For a detailed analysis of conceptual and methodological ambiguities in the measurement of frames, see Scheufele (1999).

linguistics with studies of social networks and topic modeling from sociology and computer science. Polletta is currently synthesizing linguistics and cultural sociology using new visualization techniques that enable them to explore how making people aware of their cultural schemas shapes their behavior during democratic deliberation.⁴³ Finally, Ignatow and Mihalcea (2013) propose a model for big data analysis that synthesizes neuroscience and Bourdieusian practice theory.

A final concern for cultural sociologists is the relatively high entry cost for those who wish to develop the technical expertise currently necessary to work with big data.⁴⁴ Although these costs are rapidly decreasing thanks to simple web-based tools for big data analyses, formalizing these techniques for cultural sociology will require a new generation of scholars with both technical expertise and theoretical ambition. For now, the big data movement urgently requires the guidance of theoretically and qualitatively oriented cultural sociologists. Little can be learned from big data without big thinking. While data mining may reveal interesting patterns in large text corpora or compelling visualizations, many pieces of hay have come to resemble needles.⁴⁵ Therefore, the future of the big data movement hinges upon collaboration among cultural sociologists, computer scientists, and others to teach computers to differentiate different types of meaning and their shifting relationships over time.

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⁴³ See Baumer et al. (2013)

⁴⁴ Efforts are currently underway to make the collection and analysis of big data possible for those without a computer programming background. Gary King and colleagues are producing a web-based tool named “Consilience” that will enable cluster analysis of unstructured text. Primitive forms of topic modeling and sentiment analysis are available via a variety of web-based software programs as well such as www.discovertext.com. Finally, there is a variety of high quality tutorials available online for those who wish to develop basic programming skills for working with big data. For example, see <http://nealcaren.web.unc.edu/big-data/>, and <http://www.chrisbail.net/p/software.html>. A complete list of tutorials is available at <http://www.chrisbail.net/p/big-data.html>.

⁴⁵ See Steve Lohr, “The Age of Big Data,” *The New York Times*, February 11, 2012.

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